Engineering Exploration (Laboratory)

Student Reference Manual

Module 1

Introduction to Engineering and Engineering Study



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Experiment No. 1

Title: Introduction to Engineering and Engineering Study

Objectives:

1. Explain the meaning and role of engineering in society.
2. Distinguish between science and engineering.
3. Identify engineering applications in your surroundings and the human needs they address.
4. Identify major engineering domains and the kinds of problems they address.
5. Describe the skills an engineering student should develop.
6. Identify a simple real-world problem and propose an engineering solution for it.
7. Join the GitHub organization and access your group repository.
8. Navigate the repository, edit the README and upload your work using the GitHub web interface.

Apparatus:

- A computer with internet access
- A web browser
- A GitHub account

Introduction: This module introduces you to engineering as a profession and as a way of solving real-world problems. Through observation, discussion and a group activity, you will explore the role of engineering in everyday life, the different domains of engineering, and the skills expected of an engineering student.

The module also introduces **GitHub**, a platform for storing and sharing engineering work. Each group is given access to a private repository under the COEP-EE GitHub Organization. You will use the GitHub website to access the repository, update the README and upload your Module 1 submissions.

This manual tells you **what you must do and what you must submit**. The concepts behind each activity are introduced and discussed in the laboratory and class sessions, so attend the sessions and make your own notes.

The manual has two parts:

- **Part A:** Engineering Exploration (activities and submissions)
- **Part B:** Introduction to GitHub for Engineering Collaboration

Part I

ENGINEERING EXPLORATION

1 About This Practical

1.1 What the sessions cover

The following topics are introduced and discussed during the laboratory and class sessions:

- The meaning and role of engineering
- Science and engineering, and the scientist and the engineer
- Human needs and wants
- Engineering disciplines and interdisciplinary engineering
- How engineers approach and solve problems
- Common misconceptions about engineering
- The 21st-century engineer and graduate attributes

You are expected to take notes during these sessions. You will need them for your submissions.

1.2 What you must do

1. Observe your surroundings: home, college and community.
2. Complete the **Individual Submission** on your own.
3. Work with your group on the **Group Activity**.
4. Prepare a neat, organized report for each.
5. Upload your submissions to your group GitHub repository (Part B).

2 Overview of Submissions

Submission	Who	Content	Format	Filename
Individual Submission	Each student, independently	Tasks 1 to 5	Word or PDF	`Group-1_Module-1_Individual_<UCMSID>`
Group Activity Submission	The whole group, together	Identify a problem and propose an engineering solution	Word or PDF	`Group-1_Module-1_GroupReport`

Replace `1` with your group number and `<UCMSID>` with your own UCMS ID. Filename rules are in Section 16, and upload instructions are in Section 15.

3 Individual Submission

Complete the following five tasks on your own. Use examples from your own home, college and surroundings.

3.1 Task 1: Science vs Engineering

Identify **five examples from daily life**. For each example, explain:

- The scientific principle involved
- How engineering applies that principle to solve a practical problem

3.2 Task 2: Engineering Around Us

Observe your home, college or surroundings. Identify **five engineering applications**. For each, explain the **human need** it addresses.

3.3 Task 3: Different Domains of Engineering

Select any **five engineering domains**. For each domain, mention:

- One real-world problem it addresses
- One example of a solution

3.4 Task 4: Why Engineering?

Write a short reflection of **150–200 words** on the following questions:

- Why is engineering required in society?
- What would be different in everyday life without modern engineering?

3.5 Task 5: Engineering Study

Write **three skills** you think an engineering student should develop. For each skill, explain how it will help in solving real-life problems.

Additionally, pick **one technology** from the following list (discussed in class) and write a brief, reasoned argument for why it will have the greatest impact on India by 2040:

- Artificial Intelligence (AI)
- Internet of Things (IoT)
- Robotics
- Electric Vehicles
- Renewable Energy
- 3D Printing
- Smart Manufacturing

4 Group Activity Submission

4.1 Activity: Identify a Problem and Propose an Engineering Solution

Each group should:

1. Identify **one simple problem** in the college, home or community.
2. Explain **who faces the problem** and **why it matters**.
3. Brainstorm **at least three possible solutions**.
4. Select **one solution** and draw a **labelled concept sketch**.
5. Mention the **materials, tools or engineering domains** required.
6. Explain **how the proposed solution could be tested**.

This activity stays at the level of problem identification and an initial concept. Detailed design and construction belong to later modules.

5 Instructions for Preparing Your Submissions

1. Complete the individual submission **independently**.
2. Work **together as a group** for the problem identification and solution-design activity.
3. Use examples from your **home, college or community**.
4. Include **labelled sketches, photographs or diagrams** wherever relevant.
5. Clearly explain the problem, the proposed solution and its connection to engineering.
6. Prepare a neatly organized report as described below.

5.1 Report structure

Individual report (one per student):

1. Title page with your name, roll number and group details
2. Science vs Engineering (Task 1)
3. Engineering applications in daily life (Task 2)
4. Different domains of engineering (Task 3)
5. Individual reflection (Tasks 4 and 5)

Group report (one per group):

1. Title page with the names, roll numbers and details of all group members
2. Problem identification and proposed solution (Group Activity, steps 1 to 3, 5 and 6)
3. Concept sketch (Group Activity, step 4)

Length: The suggested length is **4 to 6 pages** per student and 4 to 6 pages for the group.

5.2 Sketches and photographs

- Hand-drawn sketches are acceptable. Take a clear photograph or scan of the sketch and insert it into your document.
- Label the parts of your sketch clearly.
- Each uploaded file must be **25 MB or smaller**. If your document is too large, compress the images before saving or exporting.

6 Assessment

Your submissions will be assessed on:

- Conceptual understanding
- Observation
- Creativity
- Problem-solving approach
- Teamwork
- Clarity of presentation

Part II

INTRODUCTION TO GITHUB FOR ENGINEERING COLLABORATION

7 What is GitHub?

GitHub is a web-based platform for storing, organizing, documenting and sharing project work. In this course, GitHub is the common workspace for your student group.

In this module you will use the GitHub website to:

- Join the COEP-EE GitHub Organization
- Open your group repository and view its files
- Edit the README file
- Upload your Module 1 work to the correct folder

Everything is done in a web browser. You do **not** need to install any software or use commands.

7.1 Key terms

Term	Meaning
GitHub	A web-based platform that hosts repositories and provides tools for collaboration and version control
Repository	A project folder on GitHub that stores your files and their history
Organization	A shared GitHub account that holds many repositories. Ours is COEP-EE
README	The main information page of a repository, shown on its front page
Folder	A directory inside the repository that groups related files
Commit changes	The green button that saves your change to the repository, together with a short description of what you changed
Main branch	The main version of the repository. All your work is saved here
Issues	A tool for tracking tasks, bugs, or ideas related to your project
Wiki	A dedicated section in the repository for writing long-form project documentation
Pull Request	A request to propose and review changes before merging them into the main branch
Clone	Downloading a copy of a remote GitHub repository to your local computer
Push	Uploading your locally saved changes (commits) back to the remote GitHub repository

8 Your Group Repository

Each group receives one repository under the **COEP-EE GitHub Organization**. Repository names follow this pattern:

```
FY26-27-Sem1-EE-A-Group-1
```

Key points:

- Your repository is **private**. It can be seen only by your group members and the course faculty.
- Every member of your group has **Write access** to the repository. This means you can upload files and edit files in it.
- Your group has five members.

9 Joining the GitHub Organization

9.1 Before you begin

- Create a GitHub account at `github.com` if you do not have one.
- Make sure to use your college provided email only (ending with `coeptech.ac.in` domain) for your GitHub account. The invitation is sent to it.
- **Recommended:** Turn on two-factor authentication (2FA) in your GitHub account settings to keep your account secure.

9.2 Procedure

1. Open the email invitation from GitHub, or the notification in your GitHub account.
2. Sign in to the **same GitHub account** to which the invitation was sent.
3. Review the invitation details.
4. Select **Join COEP-EE**.
5. Open GitHub and confirm that the organization is visible in your account.
6. Open your assigned repository.



Figure 1: Organization invitation email with the Join button

Important

The invitation is valid for **7 days**. Accept it within this period. If it expires, contact your faculty to request a new invitation.

9.3 Expected outcome

```
GitHub Account
      |
      V
COEP-EE Organization
      |
      V
Assigned Group Repository
```

10 Repository Structure

Every group repository follows the same standard structure.

```
Group Repository
|
+-- README.md
|
+-- Modules/
|   +-- Module-1/
|   +-- Module-2/
|   +-- Module-3/
|   \-- Module-4/
|
+-- Projects/
|   +-- Mini-Project/
|   \-- Final-Project/
|
+-- src/
+-- assets/
+-- docs/
\-- .github/
```

Item	Purpose
`README.md`	Front page with group and repository information
`Modules/`	Work related to the course modules
`Projects/`	Mini-project and final-project material
`src/`	Source code and scripts used in projects
`assets/`	Images, diagrams and other supporting files
`docs/`	General team documentation
`.github/`	GitHub configuration files

Important

The repository covers the whole academic year. For Module 1, you only work in `README.md` and `Modules/Module-1/`. Do not modify or upload files anywhere else unless your faculty instructs you to. Folders for later modules will be created when needed, and your faculty will explain how in class.

11 The Repository README

The `README.md` file in the root of the repository is displayed on the repository's front page. It gives an overview of the group and the repository.

It contains a **team members table**:

Name	UCMS ID	GitHub Username
Member 1	UCMS ID	@username
Member 2	UCMS ID	@username
Member 3	UCMS ID	@username
Member 4	UCMS ID	@username
Member 5	UCMS ID	@username

It also contains a brief description of the repository folders. You will update the team members table in GitHub Activity 2.

12 Navigating the Repository

You should be able to:

1. Open your assigned repository.
2. View the `README.md`.
3. Open folders and files.
4. Return to the repository root by selecting the repository name at the top of the page.
5. Locate the `Modules/Module-1/` folder.
6. View the files inside it.

12.1 GitHub Activity 1: Repository Familiarization

Locate and note down:

- The repository name
- The group name
- The `README.md` file
- The `Modules/` folder
- The `Modules/Module-1/` folder, which is your Module 1 workspace

13 Editing Files on GitHub

You can edit text files, such as the README, directly in your browser.

13.1 Procedure

1. Open the file, for example `README.md`.
2. Select the **pencil icon** (Edit this file) at the top right of the file.

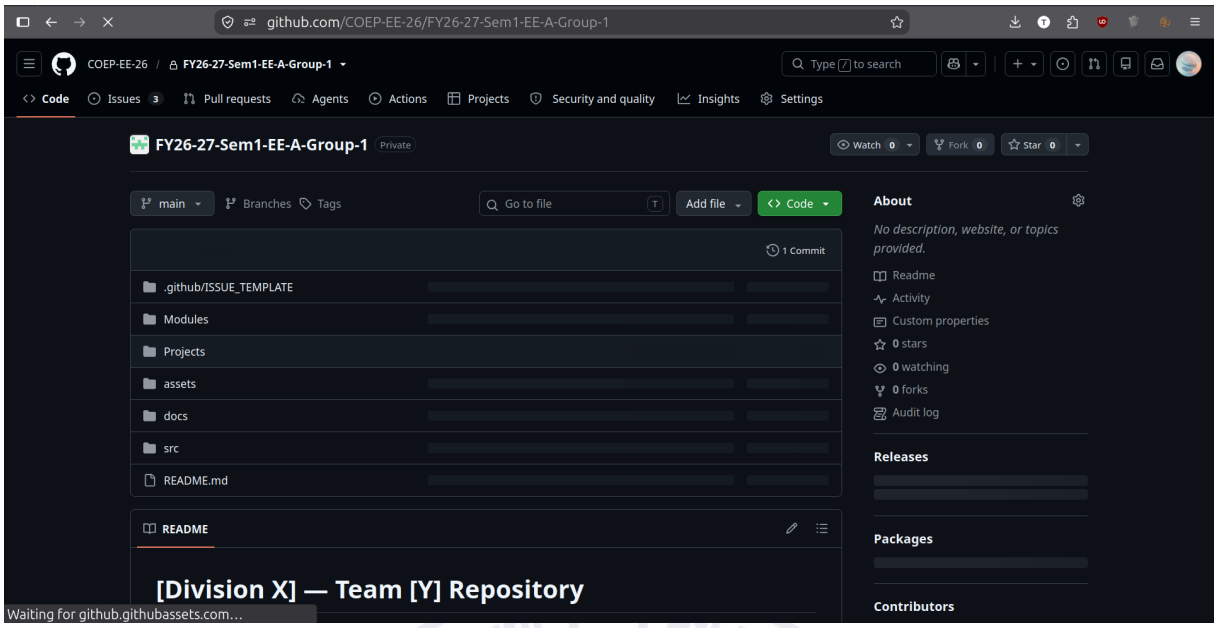


Figure 2: Repository front page showing the README, Modules folder and file structure

3. Make your changes in the editor.
4. Select the **Preview** tab to check how the page will look.
5. Select **Commit changes...**.
6. In the message box, write a short description, for example `Add team member details`.
7. Choose **Commit directly to the main branch**.
8. Select **Commit changes**.

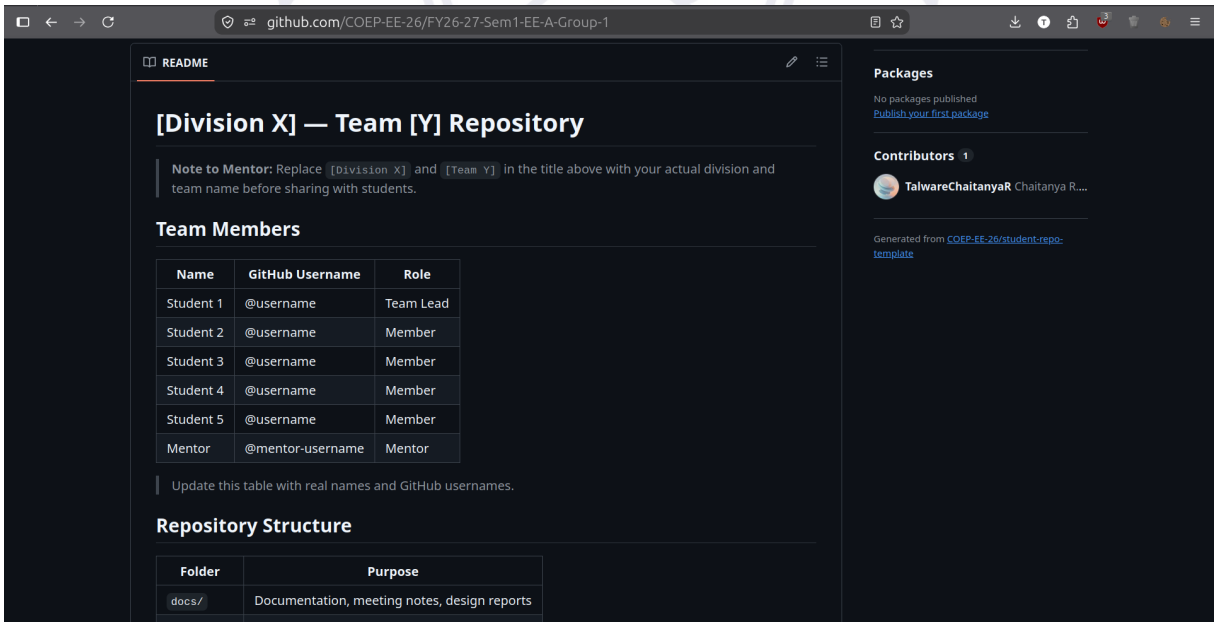


Figure 3: README section showing the pencil (edit) icon and Commit changes button

Always commit directly to the main branch. Do not create new branches or pull requests.

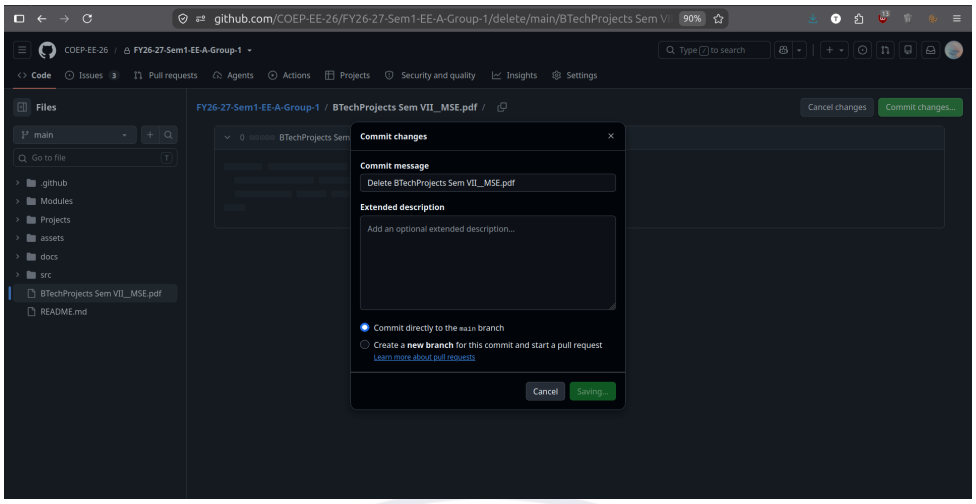


Figure 4: Commit dialog with "Commit directly to the main branch" selected

13.2 Basic Markdown

The README and other GitHub text pages are written in **Markdown**, a simple way of formatting text using plain characters. You will use it mainly for the README. Word or PDF is used for your submissions, so you do not need to learn Markdown in depth.

You type	You see
`# Heading`	A large heading
`## Smaller Heading`	A smaller heading
`**bold text**`	**bold text**
`*italic text*`	*italic text*
`- item`	A bulleted list item
`1.  item`	A numbered list item
`[COEP](https://www.coep.org.in)`	A clickable link

13.3 Markdown tables

A table is written using vertical bars and dashes:

```
| Name    | UCMS ID       | GitHub Username |
| ------- | ---------- | --------------- |
| Asha    | 112600001 | @asha-dev       |
| Ravi    | 112600002 | @ravi-codes     |
```

This displays as:

Name	UCMS ID	GitHub Username
Asha	112600001	@asha-dev
Ravi	112600002	@ravi-codes

Each row is one line. The second line (the dashes) is required. Use the Preview tab to check that the table displays correctly before you commit.

13.4 GitHub Activity 2: Update the README Team Table

One member of your group edits the README and replaces the placeholder rows with the real names, UCMS IDs and GitHub usernames of all five members.

Steps:

1. Open `README.md` and select the pencil icon.
2. Replace `Member 1`, `MIS` and `@username` in each row with the correct details.
3. Check the result using the Preview tab.
4. Commit directly to the main branch with the message `Update team members`.

14 Creating Folders and Files

Later in the course you will add your own folders, such as one for a new module. Your faculty will tell you which folders to create and when.

GitHub does not allow completely empty folders, so a folder is created together with a file inside it.

14.1 Procedure

1. Open the folder in which you want to create the new folder, for example `Modules/`.
2. Select **Add file**, then **Create new file**.
3. In the filename box, type the new folder name followed by a slash, then a file name. For example: `Module-5/README.md`. When you type the `/`, GitHub turns the text before it into a folder.
4. Add some content, for example a heading such as `# Module 5`.
5. Select **Commit changes...**, choose **Commit directly to the main branch**, and select **Commit changes**.

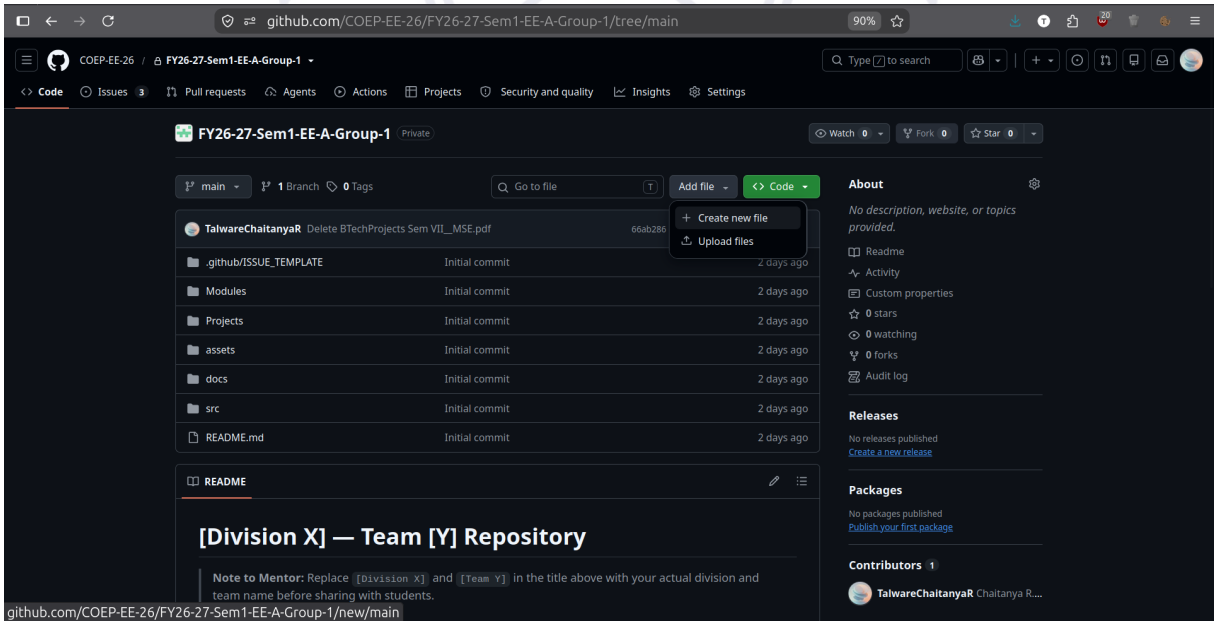


Figure 5: The "Add file" menu showing the Create new file option

15 Uploading Files

You will upload your submissions through the GitHub website.

15.1 Procedure

1. Open your repository and go to the required folder, for example `Modules/Module-1/`.
2. Select **Add file**, then **Upload files**.
3. Drag your file into the page, or select **choose your files** and pick it from your computer.
4. Check that the filename is correct.
5. In the message box, write a short description, for example `Add individual submission`.
6. Choose **Commit directly to the main branch**.
7. Select **Commit changes**.
8. Confirm that the file now appears in the folder.

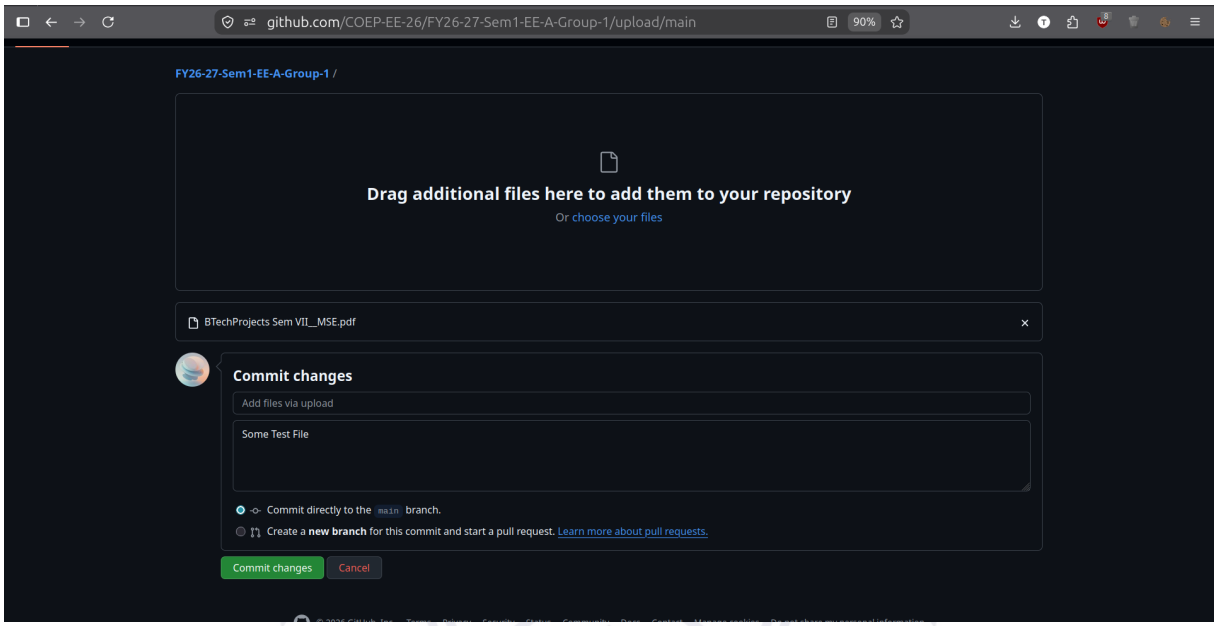


Figure 6: Upload files page with a file added and the Commit changes section visible

15.2 Workflow

```
Open Repository
      ↓
Open Required Folder
      ↓
Add file → Upload files
      ↓
Select Files and Check Filenames
      ↓
Write Description
      ↓
Commit Changes (to main branch)
      ↓
Verify File in Repository
```

15.3 File size limit

Each file uploaded through the browser must be **25 MB or smaller**. If your file is larger, reduce its size, for example by compressing the images in the document, before uploading.

16 File Naming

Use one consistent naming style for all files.

16.1 Format

```
Group-<Number>_Module-<Number>_<Description>.<extension>
```

16.2 Examples for this module

```
Group-1_Module-1_Individual_112600001.pdf
Group-1_Module-1_GroupReport.pdf
Group-1_Module-1_GroupReport.docx
```

The individual file includes your own UCMS ID.

16.3 Rules

- Replace the group number with your own.
- Do not use spaces in filenames. Use capital letters to separate words.
- Do not use vague names such as `final.pdf`, `final2.pdf` or `final_latest.pdf`.
- If you need to upload a corrected version, upload it with the same filename. GitHub keeps the history of earlier versions.

17 Repository Guidelines

17.1 You should

- Use your own GitHub account.
- Use meaningful filenames as described in Section 16.
- Follow the prescribed folder structure.
- Upload work only to the location your faculty specifies.
- Commit directly to the main branch.
- Check your files after uploading.

17.2 You should not

- Delete files uploaded by your group members.
- Change repository settings.
- Upload passwords, API keys or other confidential information.
- Upload unnecessary or unrelated personal files.
- Change files that belong to another activity or module without permission.
- Share your GitHub account credentials with anyone.

18 Module 1 Submission on GitHub

18.1 GitHub Activity 3: Upload Module 1 Work

Upload the following files (Word or PDF) to `Modules/Module-1/` in your repository:

Document	Uploaded by	Filename
Individual Submission	Each member, for their own report	`Group-1_Module-1_Individual_<UCMSID>`
Group Activity Submission	One member, for the group	`Group-1_Module-1_GroupReport`

18.2 Summary of GitHub tasks

Activity	Task	Where
GitHub Activity 1	Repository familiarization	GitHub
GitHub Activity 2	Update the README team table	`README.md`
GitHub Activity 3	Upload the Module 1 documents	`Modules/Module-1/`

18.3 Before you finish

- Join the GitHub organization
- Access the assigned repository
- Complete the repository familiarization
- Update the README team table
- Upload the Module 1 files and verify that they appear in the repository

19 Learning Check

1. What is engineering?
2. How is engineering different from science?
3. What is the difference between a scientist and an engineer?
4. What is the difference between a need and a want?
5. Name three engineering domains and one problem each addresses.
6. What are the main stages of the engineering problem-solving process?
7. Mention three misconceptions about engineering.
8. What skills are expected of a 21st-century engineer?
9. In the Target-Hitting Structure activity, why were only 2 test shots allowed before final scoring? What does this constraint teach about engineering trade-offs?
10. Which technology do you think will change India the most by 2040, and why?
11. What is GitHub?
12. What is a repository?
13. What is the purpose of a README file?
14. What does the **Commit changes** button do?
15. Why should files be named and organized according to the prescribed structure?

20 Module 1 Checklist

20.1 Engineering Exploration

- ☐ Attended the laboratory and class sessions and made notes
- ☐ Completed Task 1 (Science vs Engineering)
- ☐ Completed Task 2 (Engineering Around Us)
- ☐ Completed Task 3 (Different Domains of Engineering)

- ☐ Completed Task 4 (Why Engineering?) within 150–200 words
- ☐ Completed Task 5 (Engineering Study and Future Technology argument)
- ☐ Participated in the Target-Hitting Structure activity
- ☐ Took part in the Group Activity
- ☐ Included labelled sketches, photographs or diagrams
- ☐ Prepared the individual report with a title page
- ☐ Prepared the group report with a title page

20.2 GitHub

- ☐ Created or verified GitHub account
- ☐ Accepted organization invitation within 7 days
- ☐ Accessed assigned repository
- ☐ Read the repository README
- ☐ Located `Modules/Module-1/`
- ☐ Updated the README team table
- ☐ Uploaded the Module 1 documents (each file 25 MB or smaller)
- ☐ Verified that uploaded files appear in the repository